

Lecture 12

*Image Topology & Distance Transforms:
Estimating and Assessing Distance in Images*

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Contents

- Distance
- Distance transformations
- Binary Neighbourhood operations
- Medial Axis
- Medial Axis Transformation
- Skeleton
- Skeletonisation
- Convex Hull
- Applications

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Distance

- Distance:
 - Length of a line segment between two points, lines, planes.
- Images:
 - Distance is expressed as distance in pixels.
- Distance:
 - Shortest path between two points on the grid, c.q. two centers of pixels (in the 2D case)

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Distance Transformation

- The distance transform: a binary image is transformed such that each *pixel* of a new image represents the distance to one *pixel* or a set of pixels in the original image.
- This can be scaled to more dimensions. Thus read *imel* in stead of *pixel*.
- The distance transform converts a binary image containing an object and its background to an image where all *imels* represent the distance to the nearest background *imel*.

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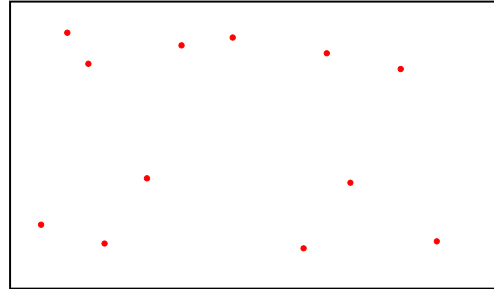
Voronoi Diagram

- The distance transform is the equivalent of a voronoi diagram in image space: or on a grid/lattice of points
- The voronoi diagram divides the a space in areas with respect to a point. All points in an area are closer to that point than to any other “defined” point in the plane.
- An example with a set of points in the plane.

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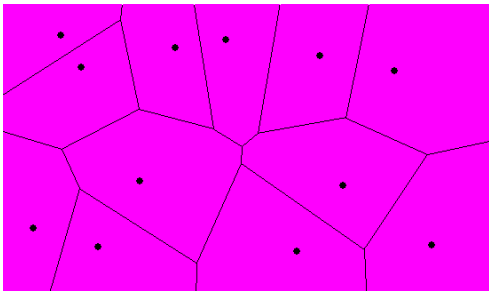
A set of points in a plane



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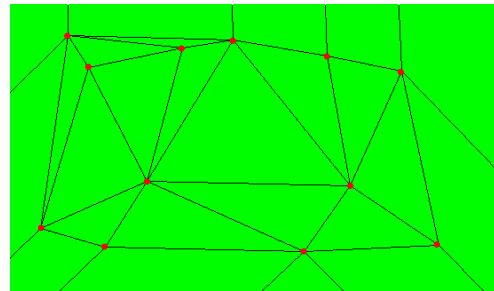
The Voronoi diagram



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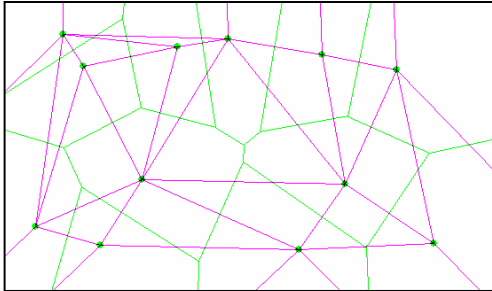
The Delaunay Triangulation



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Dual Relation Delaunay-Voronoi

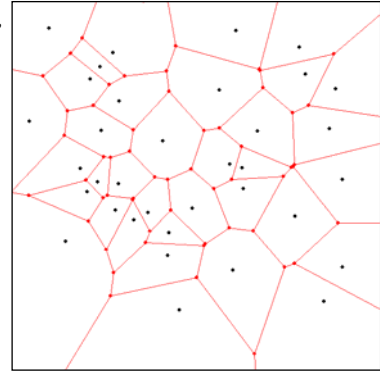


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Distances & Size Voronoi Cells

- The closer the points, the smaller the cells.
- Voronoi diagram describes continuous case of distance map.
- Computational Geometry



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Distance & Metrics

- Distance in images translates to distance on a grid or lattice of points.
- Metric:
 - Path: on a grid a path is defined as a sequence of basic vectors
 - Length: the length of the path is the sum of the distances associated with the basic vectors
 - Distance: the distance between two points is the minimum path-length between these points

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Distance Transform: metrics

- An important feature of the Distance Transform is that it represents distances on a grid. Thus a metric is required:
 - City block distance
 - Chessboard distance
 - Chamfer distance
 - Hexa-decagonal distance
 - Euclidean distance
 - Vector distance

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Metrics and Connectivity

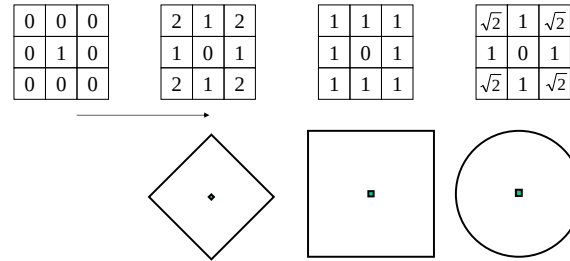
- City block distance: measures the path between pixels based on a 4 connected neighbourhood.
- Chessboard distance: measures the path between pixels based on an 8 connected neighbourhood.
- Euclidean distance: shortest distance between two pixels

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Metrics and Isodistances

- The Iso-distances show quality of the metric with respect to accuracy.



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Standard Metrics

- City Block Distance $|x_1 - x_2| + |y_1 - y_2|$
 - Taxi Cab
 - L_1
 - Manhattan
- Chessboard $\max(|x_2 - x_1|, |y_2 - y_1|)$
 - Chebychev

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Other Metrics

- Octagonal
 - iso-distances form an octagon
- Hexadecagonal
 - iso-distances form a hexadecagon

7	5	7
5	0	5
7	5	7

	11		11	
11	7	5	7	11
	5	0	5	
11	7	5	7	11
	11		11	

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Chamfer Metrics

Parallel mask

4	3	4
3	0	3
4	3	4

Recursive mask

4	3	4
3	0	3
4	3	4

Forward pass

Backward pass

- Move over image with mask or structuring element
- Mask pixel replaced with distance value
- If ambiguity, then $\min(d1, d2)$

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A simple Binary Image

0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

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Distance Transform (city block)

0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	0
0	1	2	2	2	2	2	2	1	0
0	1	2	3	3	3	3	2	1	0
0	1	2	3	4	4	3	2	1	0
0	1	2	3	3	3	3	2	1	0
0	1	2	2	2	2	2	2	1	0
0	1	1	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

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Distance Transform (chessboard)

0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	0
0	1	2	2	2	2	2	2	1	0
0	1	2	3	3	3	3	2	1	0
0	1	2	3	4	4	3	2	1	0
0	1	2	3	3	3	3	2	1	0
0	1	2	2	2	2	2	2	1	0
0	1	1	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

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Very Simple Image

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

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Manhattan (City Block) Distance

8	7	6	5	4	3	2	1	0	1	2	3	4	5	6	7	8	9
7	6	5	4	3	2	1	0	1	2	3	4	5	6	7	8	9	10
6	5	4	3	2	1	0	1	2	3	4	5	6	7	8	9	10	11
5	4	3	2	1	0	1	2	3	4	5	6	7	8	9	10	11	12
4	3	2	1	0	1	2	3	4	5	6	7	8	9	10	11	12	13
3	2	1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
2	1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

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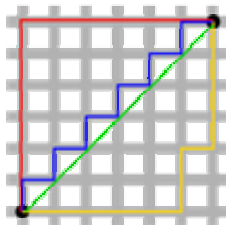


Image by Tristan Miller

— Steps: 12
 — Steps: 12
 — Steps: 12
 — Steps: $\sqrt{(6)^2 + (6)^2} = \sqrt{72} \approx 8.4853$

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Key papers Distance Transforms

- 1968, Rozenfeld & Pfaltz;
 - Distance functions on digital pictures
- 1980, Danielson;
 - Euclidean Distance Mapping
- 1986, Borgefors
 - Distance transformations in digital images
- 1990, Verwer
 - Shading from shape, grey-weighted distance transform

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Algorithmic approach

- Computationally efficient
 - Parallel
 - Recursive
 - Breath first, Depth first
 - Uniform cost
- Optimized estimators for Euclidian distance on a grid
- Minimizing the error introduced by metric

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Relation Mathematical Morphology

- Erosion can be accomplished by thresholding the distance transform of the object
 - Effect of applied metric : Structuring element!
- Dilation can be accomplished by thresholding the distance transform of the background
- This enabled a number of fast applications
 - Skeleton by thinning

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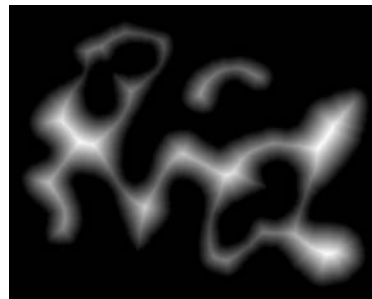
A Complex Shape



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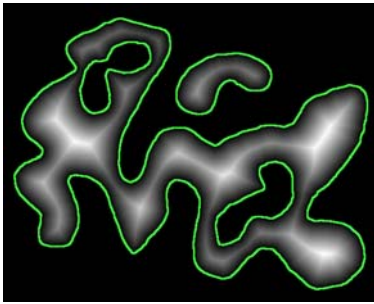
Euclidean Distance Map



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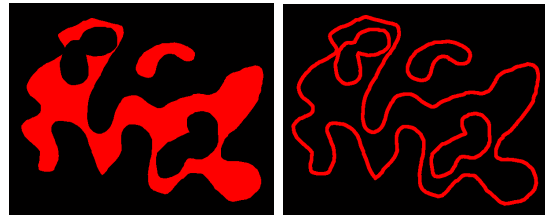
Relation with Shape



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Erosion with Distance Map



Threshold DT 9

Xor with original (difference)

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Vector Distance Transform

- Euclidean distance is constructed from a path vector.
- The length of the path-vector is the distance value at a pixel location.
- Sometimes, for convenience, $(\text{Length})^2$ is used: memory requirements
- The vector components (x,y) or (x,y,z) are also used:
 - signed Euclidean distance transform
 - vector distance transform

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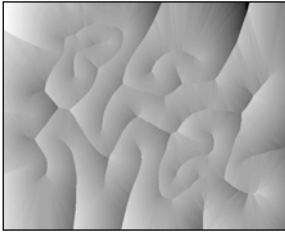
Vector Distance Map

- The VDT can be formalized as
- $EDT_{xy} = \sqrt{(V_x)^2 + (V_y)^2}$
: square/square root of images (vx and vy)
- $EDT_{xyz} = \sqrt{(V_x)^2 + (V_y)^2 + (V_z)^2}$
: square/square root of images (vx, vy and vz)

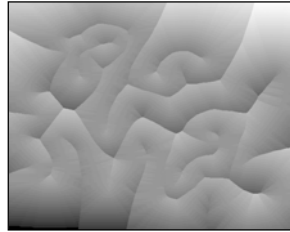
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Application of VDT



X component of the vector



Y component of the vector

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Distance map of Shape



- Contour
- Inverse
- EDT
- D(Point, Shape)

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Application Distance Map

- Registration, find a minimum in the distance map of a shape: rigid transforms (R,T)
- Dual of VDT and EDT; Multi-resolution
- Skeleton
- Watershed transformations
- Shape description

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Artistic application



Coin design by P. Struyken
adapted from work by
Dr. B. Verwer (TU-Delft)

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Medial Axis

- The medial axis is the preservation of a shape by a midline that reflects topology
- Analogy to a grass-fire
 - Quenching line = MA and Skeleton
- Medial axis is not per se defined on a grid.
- Grid: skeleton

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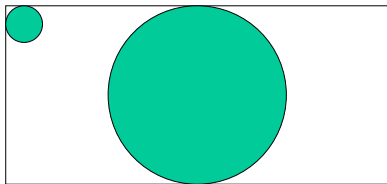
Medial Axis Transformation

- Several approaches are described.
- Blum's Medial Axis Transformation
- a.k.a. Blum's Symmetry axis
- tangent circles
 - entirely within the object
 - maximal disk at more than one point
- A MA can be used to reconstruct the shape.
 - How?
 - What information is required ?
- Linear and parabolic segments are used in more elaborate versions of Medial Axis

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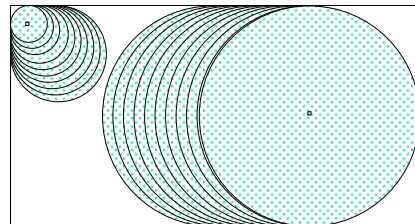
Medial Axis & Maximal Disk



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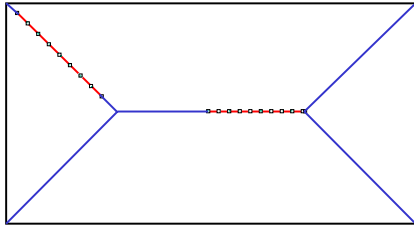
Medial Axis by Maximal Disks



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Centers of Disks Span Medial Axis



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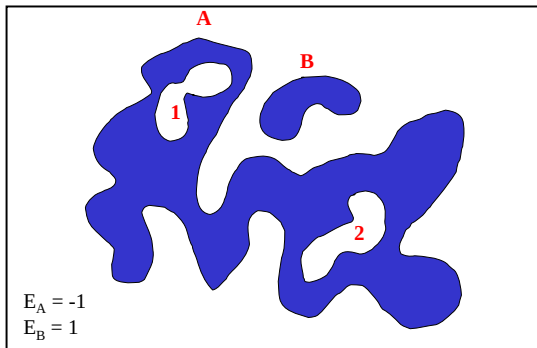
Topology

- Topology is the study of object properties that are unaffected by deformation.
- Topology is described by its Euler number.
 - let C represent a *connected region*
 - let H represent # of *holes* in a connected region
 - Then $E = C - H$ represents the Euler Number
- Euler number E is a topological descriptor.
- If a topology is preserved then the Euler number is preserved.

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Euler Number & Topology



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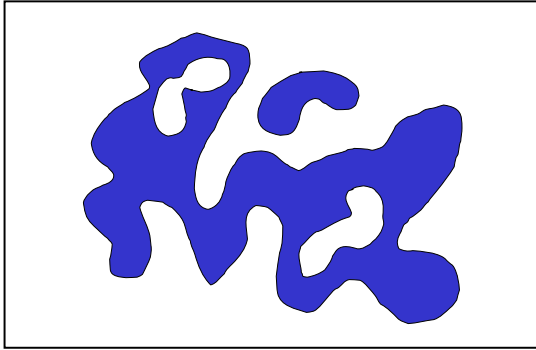
Convex Hull

- Objects, or regions have parts that are concave and parts that are convex
 - The convex hull is the minimal shape derived from a shape that is convex
- OR
- The *convex hull* H of an arbitrary set S is the smallest convex set containing S .
- ALSO
- $D = H - S$ is referred to as the *convex deficiency*. This can be measured several ways e.g. Perimeter ratio, Area ratio

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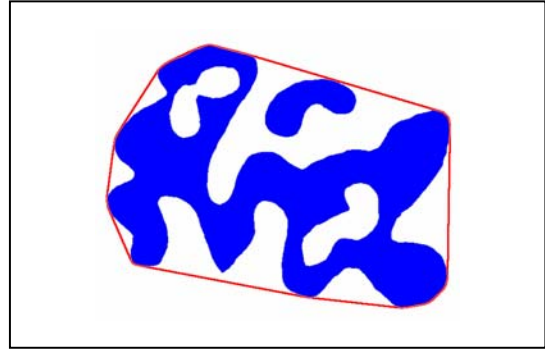
Convex Hull and Topology



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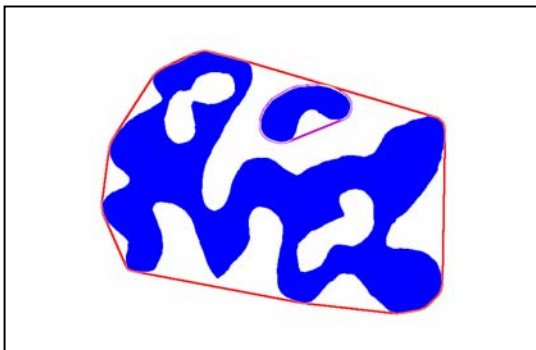
Convex Hull of all shapes



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Convex Hull of each Shape



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Skeleton

- The skeleton is the medial axis of a shape that is discretized on a grid.
 - An object resulting from a segmentation
 - Morphological skeleton: imperfections of grid
- The skeleton of an object is the set of connected pixels in the middle of the original shape representing the topology of the object.
End-pixels (protrusions) are essential part of the topology and therefore also preserved.

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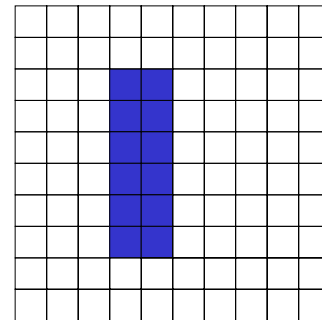
Computing the Skeleton

- The skeleton can be computed from the distance transform by thinning: or erosion.
 - One pixel thick line
 - Preserves topology
 - Pixels are removed from the object if these conditions are true
- The local maxima of the distance transform are seen as ridges. The locus of these ridges corresponds with the medial axis

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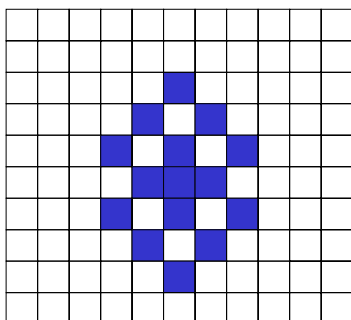
Pathological Case (1)



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Pathological Case (2)



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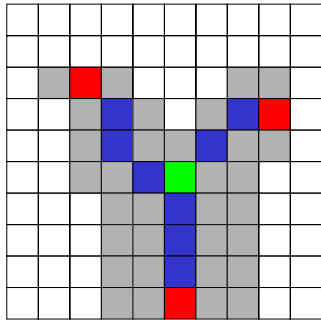
Skeleton-points

- In the computation of the skeleton several typical points are classified:
 - Skeleton point
 - Two neighbours
 - Branch point
 - Three neighbours
 - End point
 - One neighbour
 - These points are stored separately

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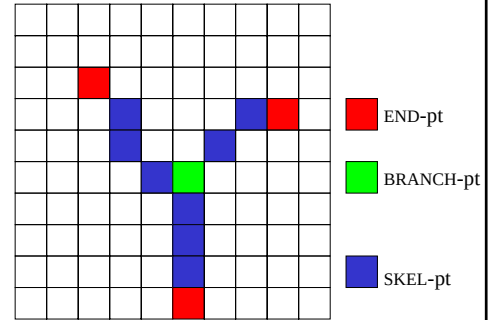
Thinning to Skeleton



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Visualization Skeleton Points



END-pt
BRANCH-pt
SKEL-pt

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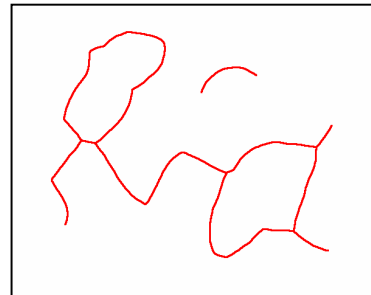
Complex Shape



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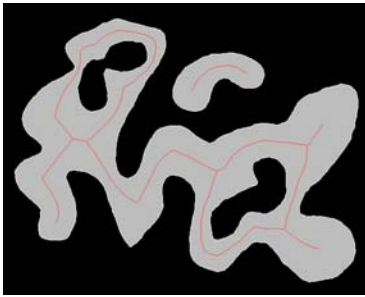
Skeleton from Euclidean DT



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Topology is preserved



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Skeleton located at local Maxima



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Exo-Skeleton

- The exo-skeleton is the skeleton of the background.
- The exo-skeleton a.k.a. [skiz](#), or geodesic skiz
- SKIZ = SKeleton Influence Zone

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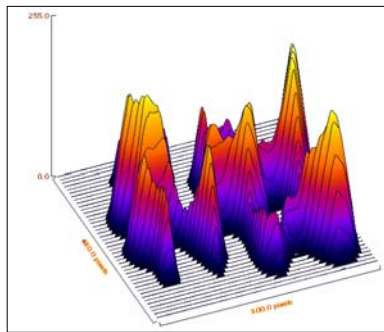
Watershed Segmentation

- SKIZ shows watershed in an image.
- Principle from geography (cartography)
- Partitioning tool based on mathematical morphology (Vincent, L. & Soille, P. 1991)
- Based on idea that grey values represents elevation in a landscape.
- illustrated with the Distance transform

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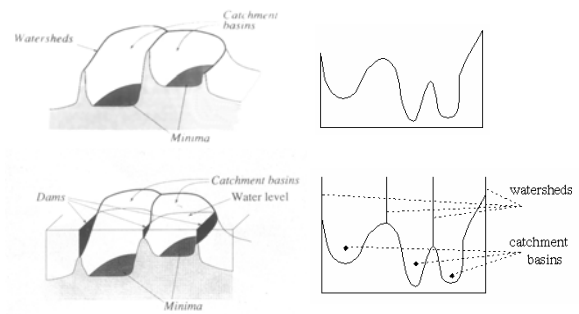
Elevation Maps of an Image



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Watersheds & Basins



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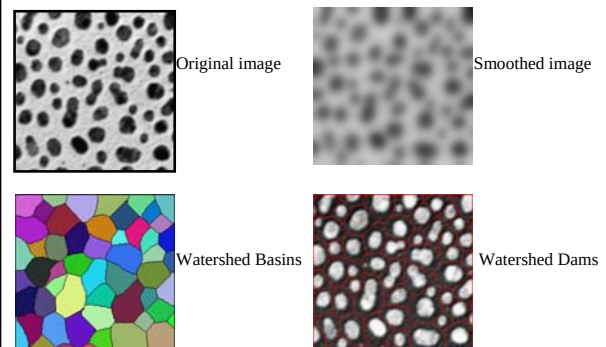
Watershed basic Algorithm

- Image data = topographic surface.
 - Gradient image gray-levels represent altitudes.
 - Region edges correspond to high watersheds
 - Low-gradient region interiors correspond to catchment basins.
- Catchment basins of the topographic surface are homogeneous
 - pixels belonging to the same catchment basin connected with the basin's region of minimum altitude
 - Using *simple path* of pixels with monotonically decreasing altitude along the path.
- Catchment basins represent the regions of the segmented image.
- **Immersion** is used to find the Dams watersheds.
- Flooding from one basin to another is prevented by putting up a dam.
- Potential disadvantage of oversegmentation.
- Depends on the strategy in segmentation (e.g. smoothing)

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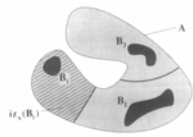
Watershed Segmentation Example



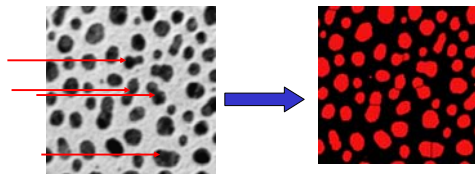
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Watershed & SKIZ



- Maxima of the distance transform
- Centers of influence
- Compare with Basins
- Maxima considered as Minima



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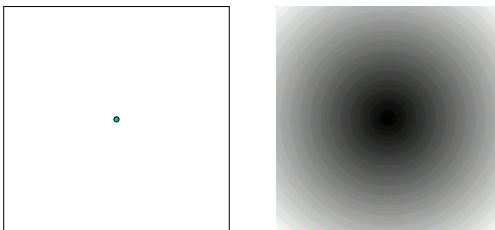
Defining Circles with EDT

- Distance transforms
 - Object
 - Background
- How can we exploit this to define a circle:
 - Point in the image, $(0.5 \cdot s_x - 1), (0.5 \cdot s_y - 1)$
 - Distance map to that point
 - Threshold at R or R^2

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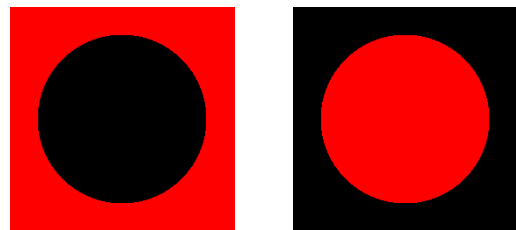
EDT of a Point (imel)



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Circle from Distance Transform



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Review Lecture 12



- Distance in images
- Distance Transformation
- Medial Axis
- Medial Axis transformation
- Skeleton
- Convex Hull
- Applications

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Acknowledgements

Books

- Gonzales & Woods: Digital Image Processing

- www.cs.cornell.edu (chew, computational geometry)

Papers

- Borgefors, G. (1986) Distance Transforms in Digital Images
- Vincent L., Soille, P. (1991) Watersheds in digital spaces: An efficient algorithm based on immersion simulations.
- Thesis Dr. Ir. Ben Verwer
- Thesis FJV, test and experimental data

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